

Retail Consumer Behavior Analysis

Detailed Business Problem Statement & Project Roadmap

1. Business Background

A leading retail company wants to understand how and why customers make purchasing decisions. The company sells products through online and offline channels. Management has noticed that customer behavior changes based on factors such as age group, product category, discounts, reviews, season, and payment method.

The company already has customer and shopping data, but it is not being used to its full potential. The business needs a clear analysis that converts raw data into useful insights that managers can understand and act upon.

2. Business Problem

The main problem is that the company does not have a clear picture of which customer groups, products, shopping channels, and purchasing factors contribute most to sales, satisfaction, and repeat purchases.

Management therefore needs a data-driven approach to identify important patterns in customer shopping behavior and use the findings to improve marketing, product planning, customer engagement, and long-term loyalty.

Main Business Question: How can the company use consumer shopping data to identify important trends, understand customer behavior, improve engagement, and make better marketing and product decisions?

3. Objectives of the Project

- Understand how customers behave across different demographic groups.
- Find the product categories that receive the highest customer interest and purchases.
- Compare online and offline shopping behavior.
- Measure the effect of discounts on purchasing decisions.
- Understand whether customer reviews or ratings are related to buying behavior.
- Study seasonal changes in shopping patterns.
- Analyze preferred payment methods and their relationship with purchases.
- Identify customer groups with stronger loyalty or repeat-purchase behavior.
- Find useful patterns that can support marketing campaigns and product decisions.
- Present findings in a simple dashboard so business stakeholders can make decisions quickly.

4. Key Business Questions

Area	Business Question
Customer Segments	Which age, gender, or demographic groups purchase most frequently or generate higher sales?
Products	Which product categories and individual products are most popular? Which categories may need improvement?
Sales Channel	How does customer behavior differ between online and offline purchases?
Discounts	Do discounts increase purchases? Which customer groups respond most strongly to discounts?
Reviews	Are better reviews or ratings associated with higher purchasing activity?
Seasonality	Which seasons or periods show higher or lower shopping activity?
Payment	Which payment methods are most preferred, and do preferences differ by customer segment?
Loyalty	What characteristics are common among repeat customers or loyal customers?
Marketing	Which customer groups should receive targeted offers or campaigns?

5. Project Scope

The project will cover four main analytical areas:

A. Customer Analysis

Study customer demographics, shopping frequency, spending behavior, and loyalty patterns.

B. Product & Purchase Analysis

Analyze product categories, purchase frequency, customer preferences, and factors that may influence the decision to buy.

C. Channel & Transaction Analysis

Compare online and offline shopping activity and organize the data into a transaction-oriented structure for SQL analysis.

D. Business Insight & Decision Support

Turn the analysis into practical recommendations for marketing, customer engagement, promotions, and product strategy.

6. Data Preparation & Modeling — Python

Python will be used to clean, transform, and explore the raw consumer behavior dataset before business analysis begins.

- Load the raw dataset and understand the available columns and data types.
- Check for missing, duplicate, or inconsistent records.
- Standardize text values such as categories, payment methods, and sales channels.
- Convert columns into appropriate formats, especially dates and numerical fields.
- Check unusual or extreme values that may affect analysis.
- Create useful calculated fields where required.
- Perform exploratory data analysis to identify distributions, patterns, and relationships.
- Prepare a clean dataset for SQL analysis and Power BI reporting.

Expected Python Output

A clean, analysis-ready dataset plus Python scripts or a notebook showing the cleaning, transformation, and exploratory analysis steps.

7. Data Analysis — SQL

SQL will be used to organize the cleaned data and answer business questions about customers, products, loyalty, purchasing patterns, and possible purchase drivers.

- Create logical tables or a structured transaction model.
- Use SELECT, WHERE, GROUP BY, ORDER BY, CASE, and aggregate functions.
- Use JOINS when information is stored across related tables.
- Calculate customer-level and product-level metrics.
- Compare customer segments and shopping channels.
- Analyze repeat purchasing and loyalty patterns.
- Study discounts, reviews, seasons, payment methods, and purchases.
- Write clear SQL queries that directly answer the business questions.

Expected SQL Output

A collection of well-organized SQL queries and result tables that provide measurable answers to the main business questions.

8. Visualization & Dashboard — Power BI

Power BI will be used to create an interactive dashboard that allows management to quickly understand customer behavior and important trends.

Executive Overview

KPIs, major trends, and overall customer/purchase performance.

Customer Demographics

Customer distribution and purchasing behavior by relevant demographic groups.

Product Performance

Top product categories, popular products, and weaker-performing areas.

Online vs. Offline

Comparison of shopping behavior and purchasing activity across sales channels.

Discount & Review Analysis

Patterns showing how discounts and reviews/ratings relate to purchasing behavior.

Seasonal Trends

Changes in shopping activity across seasons or available time periods.

Payment Preferences

Most-used payment methods and differences across customer segments.

Loyalty & Repeat Purchase

Indicators of repeat behavior and customer groups that appear more loyal.

Recommended Dashboard Design

Keep the dashboard simple and business-focused. Use KPI cards for important numbers, charts for trends and comparisons, slicers for filtering, and clear titles so a non-technical manager can understand the dashboard easily.

9. Business Insights to Look For

- Which customer segment is the most valuable or most active?
- Which products or categories drive the most purchasing activity?
- Is online shopping stronger than offline shopping, or vice versa?
- Do discounts appear to encourage purchases or mainly reduce price?
- Do highly rated products show stronger purchase activity?
- Are there seasonal periods where marketing activity should be increased?
- Which payment methods are most common among active customers?
- What patterns are shared by repeat customers?
- Which customer groups could benefit from personalized offers?
- Are there product categories or customer segments that need further investigation?

10. Expected Business Recommendations

Recommendations should be based on actual findings from the dataset rather than assumptions. Possible recommendation areas include:

- Create targeted marketing campaigns for high-value or high-potential customer segments.
- Use discounts strategically for customer groups that respond positively to promotions.
- Improve product or category strategy based on demand and customer feedback patterns.
- Strengthen the better-performing sales channel while identifying improvement opportunities in the other channel.
- Use seasonal trends to plan campaigns, inventory, and promotional activities.
- Encourage repeat purchases through suitable loyalty or retention strategies.
- Make payment experiences easier by supporting commonly preferred payment methods.
- Use customer reviews and feedback to identify products that need improvement.

11. Final Deliverables

Deliverable	What It Should Contain
1. Python Analysis	Cleaned dataset, transformation steps, exploratory analysis, and Python scripts/notebook.
2. SQL Analysis	Structured tables/model and SQL queries answering customer, product, loyalty, and purchase-driver questions.
3. Power BI Dashboard	Interactive dashboard with KPIs, trends, comparisons, filters, and key customer insights.
4. Project Report	Business problem, methodology, findings, insights, limitations, and actionable recommendations.
5. Presentation	Short visual presentation explaining the problem, approach, major findings, and recommendations.
6. GitHub Repository	Well-organized Python files, SQL scripts, dashboard files, documentation, and supporting materials.

12. Suggested End-to-End Workflow

Raw Dataset → Data Understanding → Data Cleaning → Data Transformation → Exploratory Analysis → SQL Data Modeling → SQL Business Queries → Power BI Dashboard → Business Insights → Recommendations → Report & Presentation → GitHub Repository

13. Success Criteria

The project will be successful when the raw consumer behavior data is converted into a clean, structured, and understandable analytical solution; the major business questions are answered with evidence; the Power BI dashboard communicates the findings clearly; and the final recommendations give management practical areas for improving customer engagement, marketing, products, and loyalty.